GIS Programming and Customization
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LA School Earthquake Risk Report

Introduction.

As we know Los Angeles County is located on the earthquake faults. The most powerful earthquake can reach as much as 8 magnitudes that threaten all residencies in Los Angeles. Schools are one of the most important places of priority for earthquake preparedness because they are places of learning and safety for children. In fact, many schools are older and not matched to current building safety requirements. Also, there are numerous schools that exist in Los Angeles County because of the large city population.

Therefore, the goal of this project is to prioritize earthquake safety for schools in Los Angeles county based on proximity to earthquake faults. This would help us prevent and reduce property damage and loss of life.

Figure1. Study Area: Los Angeles County, CA

Study area.

The area of this study is Los Angeles County, California. Los Angeles is one of the largest cities located in Southern California. It is the center of the nation's film and television industry. However, there are active earthquake faults under this city that threaten the safety of populations. Notice the primary risk of earthquakes came from the San Andreas faults which pass through the county.
Look at the basic map here. The area shaded in pink represents Los Angeles County. And the purple line represents earthquake faults provided by USGS.

Data and Data Processing.

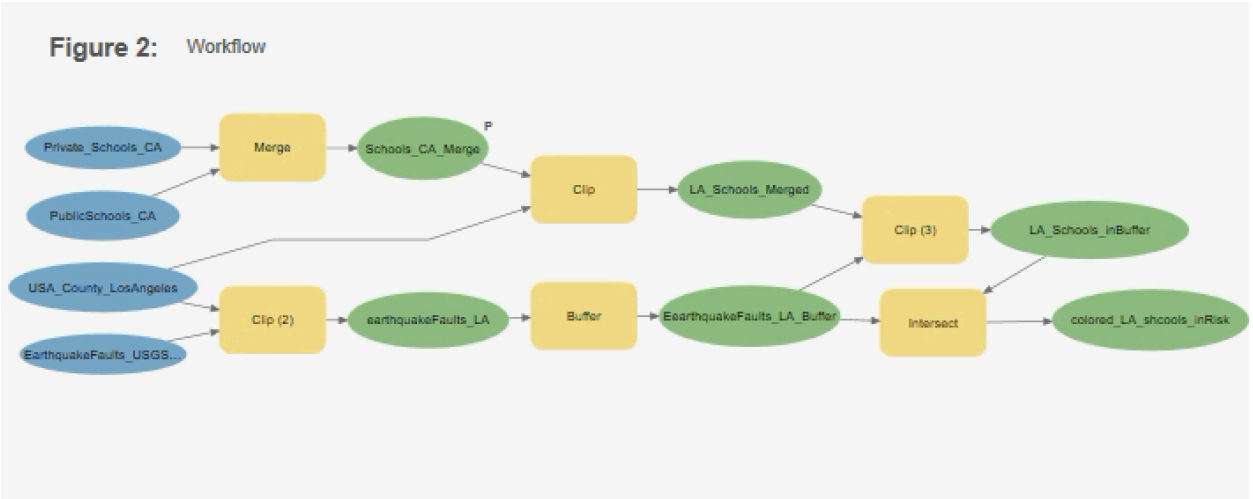
The projected coordinates of this study is in the WGS 1984 Web Mercator, the projection of Mercator Auxiliary Sphere, and the geographic coordinate system of WGS 1984. This study contains five data layers: the boundary layer of California state, the area of Los Angeles County, the distribution of earthquake faults(provided by the USGS), and the location of public and private schools.

Methods.

I use ModelBuilder in ArcGISPro to create my workflow. First, I merge the private school and public school together to get the schools merged layer since the earthquake risk to schools is equivalent whether they are public or private. Then, both schools merged layer and earthquake fault layers clipped to Los Angeles County Boundary. This step helps us visualize all schools and earthquake faults within Los Angeles County.

As the study parameters indicated that schools within 1 mile of an earthquake fault were to be identified, I used the buffer tool to create the Buffer polygon layer of 1 mile to fault lines within LA county. Next, schools merged layers clipped to the buffer to identify all LA schools in Buffer. Afterward, the intersect tool helps us get the result that all LA schools in earthquake faults Buffer.

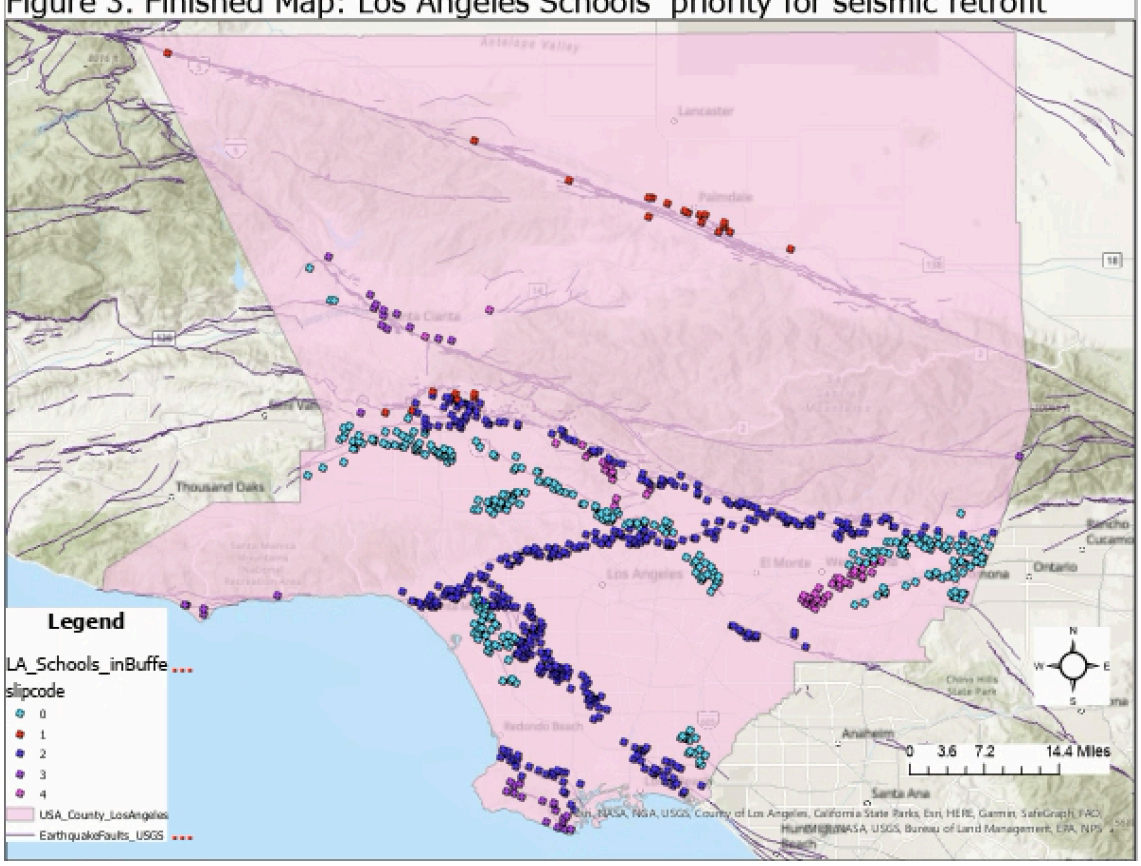
According to the information from textbooks and USGS, a positive correlation exists between the earthquake slip rate and active faults. To determine the schools that were closest to the highest risk faults and the greatest priority for seismic retrofits, symbolized this layer to graduated colors by slip code variable from 0-5. The scale of data is the slipcode because the information provided by USGS mentioned the faults with the highest slip rates have a code in 1, and the lowest have a code in 4. The faults with unknown slip rates are coded in 0. Therefore, the final layer was symbolized with the schools being color-coded based on the slip code of the earthquake fault within 1 mile.



Results.

The result of the study identified the schools within 1 mile of earthquake faults within Los Angeles County and categorized them based on the risk of the fault nearby (results shown in figure 3). The information provided by USGS mentioned the faults with the highest slip rates have a code in 1, and the lowest have a code in 4. The faults with unknown slip rates are coded in 0. And the different risk and priority levels of schools are appropriately presented with different colors shown on the map. Therefore, the highest priority and most risky schools have the slip code of 1 and the color in red.

Figure 3. Finished Map: Los Angeles Schools priority for seismic retrofit



Besides the visualization of schools under different risk levels, the code of slip rate shown in the data table is sufficient information to identify schools that should be the highest priority for seismic retrofits.

Discussion.

Assessing my model of earthquake risk analysis, there exist some limitations which can be improved in the future. Since I merge the private school and public school together to get the schools merged layer since the earthquake risk to schools is equivalent whether they are public or private. The first limitation would be that this model can not identify the distribution of private or public schools under different priority for seismic retrofits.

Moreover, I use the intersect tool to combine the earthquake faults in LA buffer layer and LA school in buffer layer to get the result that all LA schools in earthquake risk layer. Then I change the symbology to colored schools under different risk levels by slip code. This is one of the correct ways to reach the goal of this project. However, I realize there is another way to achieve the same purpose after discussing the workflow with classmates. To determine the schools that were closest to the highest risk faults, the Spatial Join tool was also useful to the model.